



Data Analysis of Software Developer Skills in Demand

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OUTLINE



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- Discussion
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EXECUTIVE SUMMARY



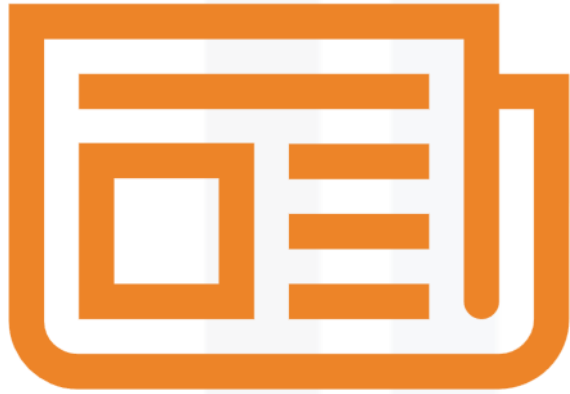
- Analyse the technology skills used and identify emerging technologies
- Analyse developer demographics such as gender, age, education and geographic location

INTRODUCTION



- What are the top programming languages in demand?
- What are the top database skills in demand?
- What are the popular IDEs?
- Are there any new technologies getting popular?
- How does developer demographics (age, gender, education level) look like?

METHODOLOGY



- Collect the top programming skills that are in demand from various sources including:
 - Job postings
 - Training portals
 - Surveys
- Clean the collected data using data wrangling techniques
- Apply statistical techniques to identify the skills on demand and analyze demographic data
- Present the findings using visualization like charts, plots and histograms

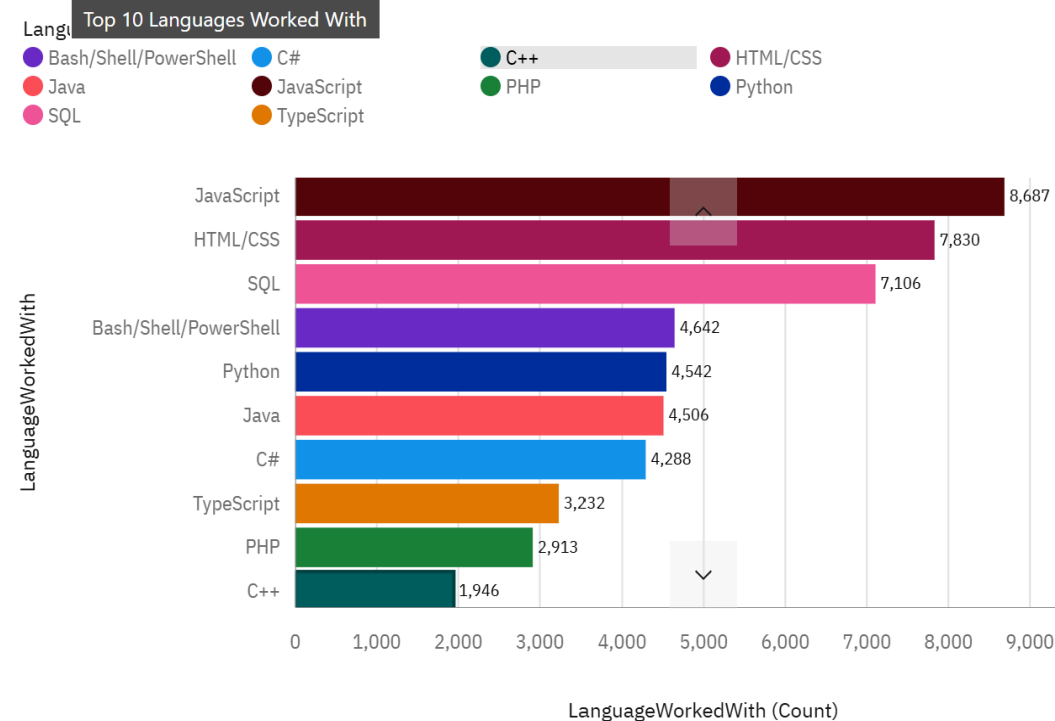
RESULTS



PROGRAMMING LANGUAGE TRENDS

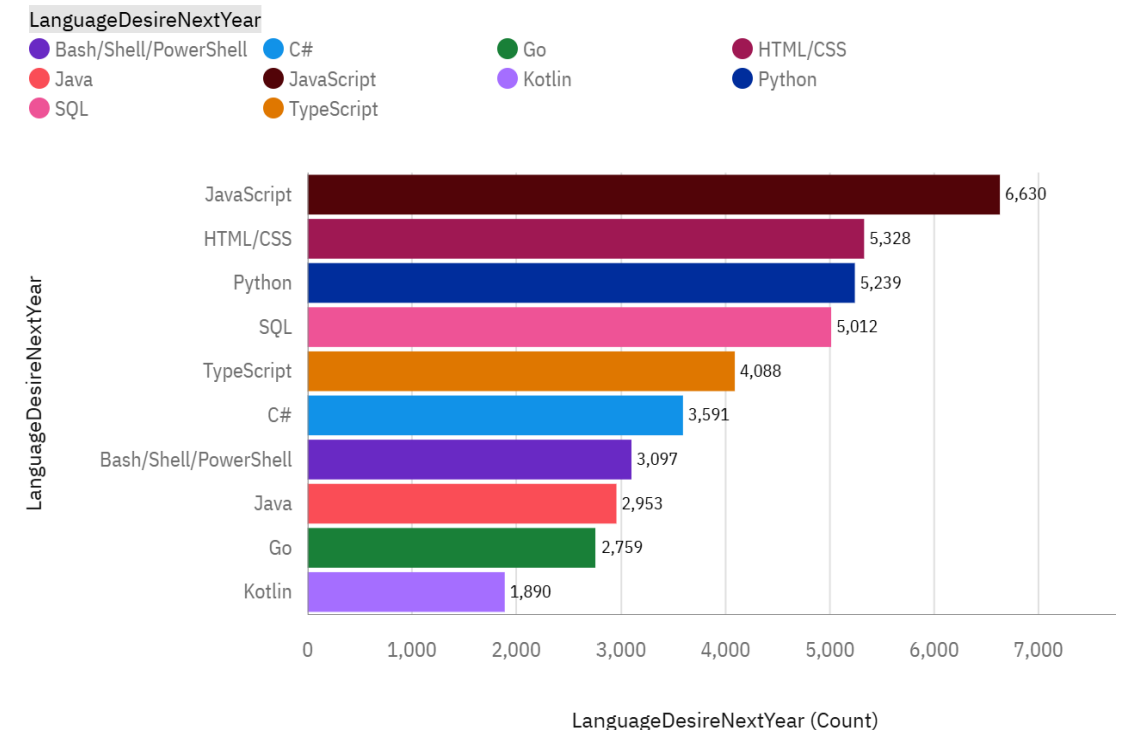
Current Year

Top 10 Languages Worked With



Next Year

Top 10 Languages Desired Next Year



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- Javascript and Html are the most popular languages used currently and also the most desired language for next year.
- Python is currently in 5th place but it is 3rd for desired languages for next year.
- Go and Kotlin are not in top 10 languages currently however more for desired languages for next there are at 9th and 10th place.

Implications

- Javascript and Html are the most widely used languages currently. Many developers are also keen to learn them in the next year.
- Python is becoming more popular as more developers showing interest to learn it next year.
- Go and Kotlin are gaining popularity with it being in the top 10 desired languages for next year.

DATABASE TRENDS

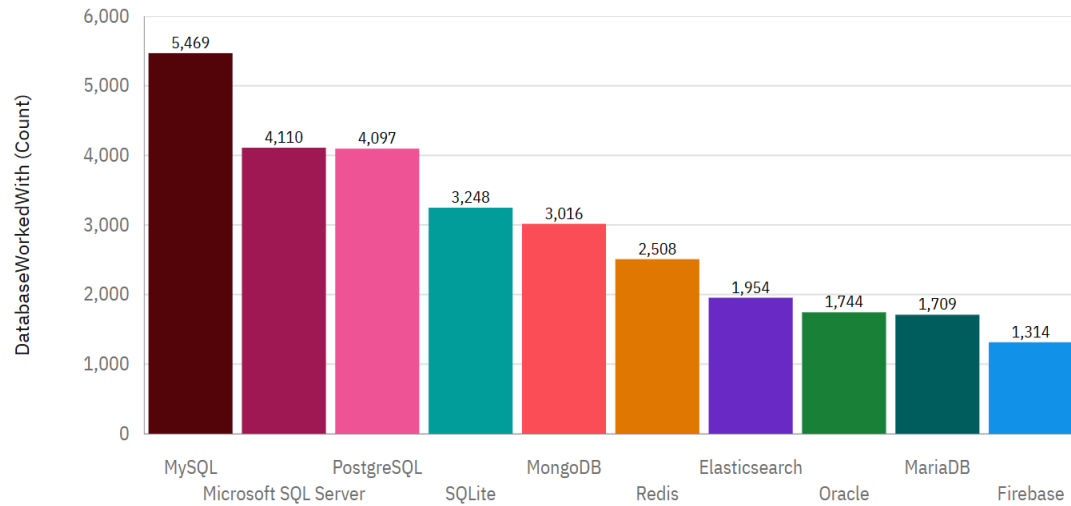
Current Year

Top 10 Database Worked With



DatabaseWorkedWith

Elasticsearch Firebase MariaDB Microsoft SQL Server MongoDB
MySQL Oracle PostgreSQL Redis SQLite



DatabaseWorkedWith

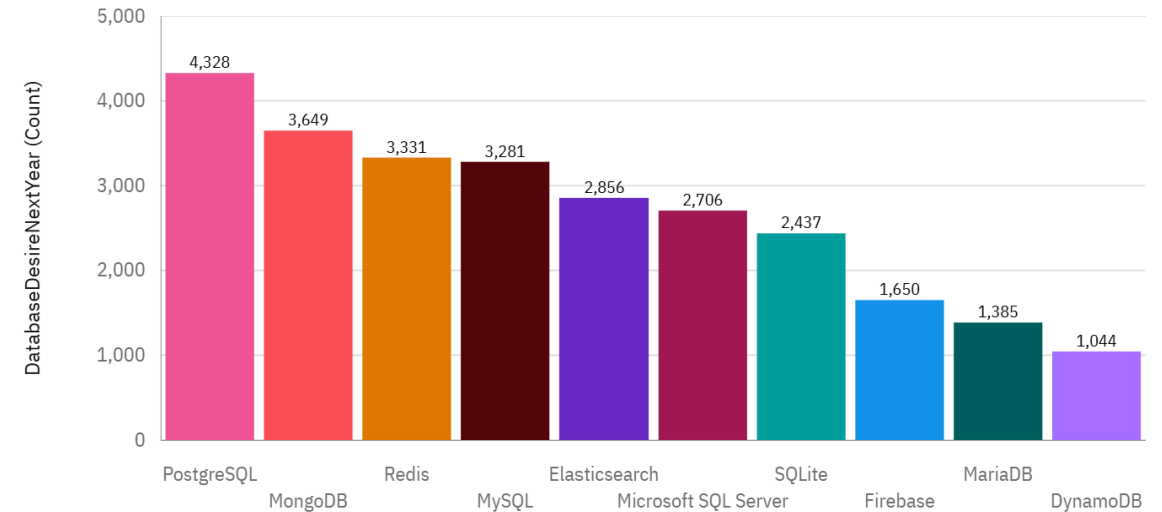
Next Year

Top 10 Databases Desired Next Year



DatabaseDesireNextYear

DynamoDB Elasticsearch Firebase MariaDB Microsoft SQL Server
MongoDB MySQL PostgreSQL Redis SQLite



DatabaseDesireNextYear

DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- MySQL and MicrosoftSQL server are the most popular Databases used currently
- PostgreSQL, MongoDB, Redis are the more desired Databases for next year
- Dynamic DB which is not in the top 10 currently is the 10th place for the next year

Implications

- There is a clear shift in the popular databases used. More people are wanting to learn PostgreSQL, MongoDB and Redis than MySql and MicrosoftSQL server which are popular now
- Dynamic Db is gaining popularity

DASHBOARD

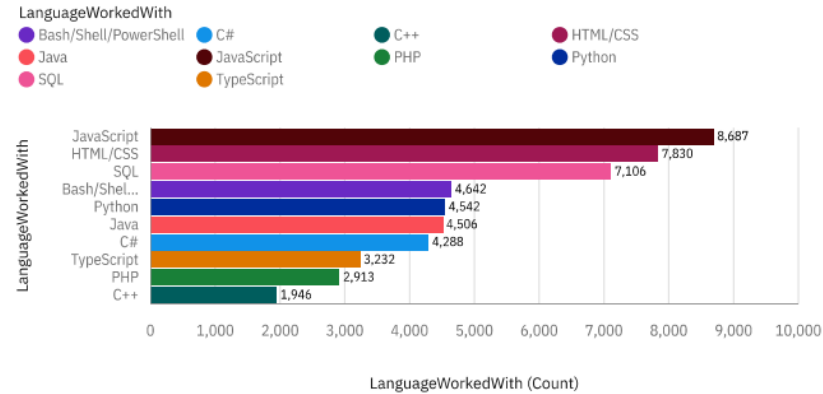


<The GitHub link of the Cognos/Looker Studio dashboard goes here.>

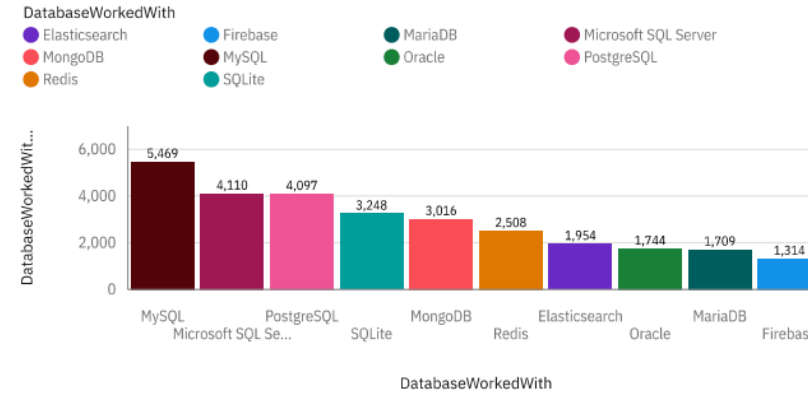
DASHBOARD TAB 1

Current Technology Usage

Top 10 Languages Worked With



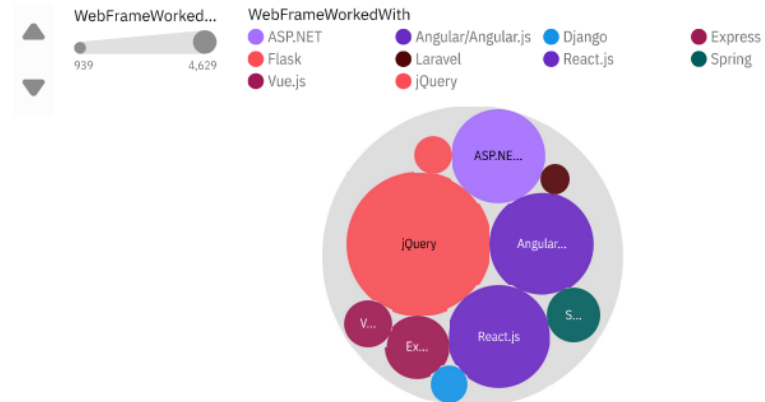
1 Top 10 Database Worked With



Platforms Worked With



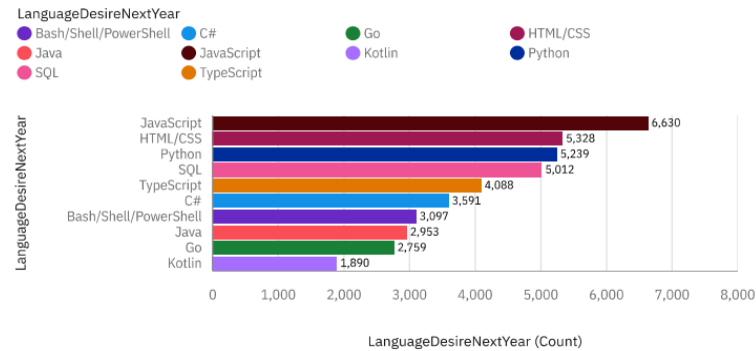
Top 10 WebFrames Worked With.



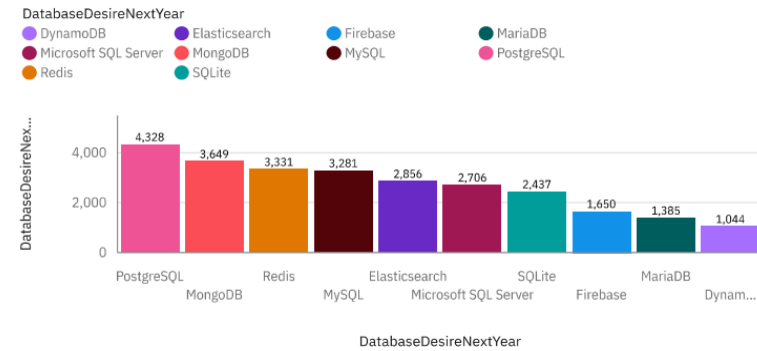
DASHBOARD TAB 2

Future Technology Trend

Top 10 Languages Desired Next Year

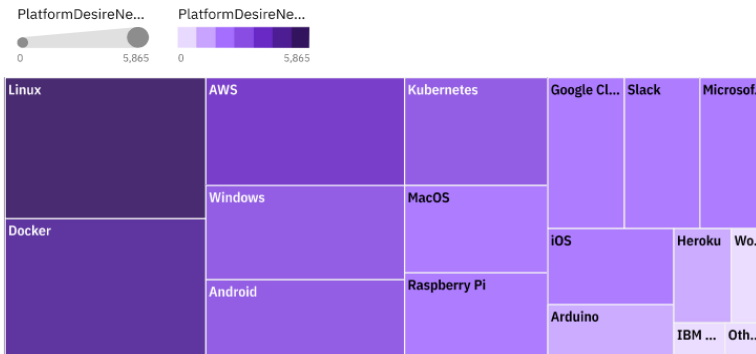


4 Top 10 Databases Desired Next Year

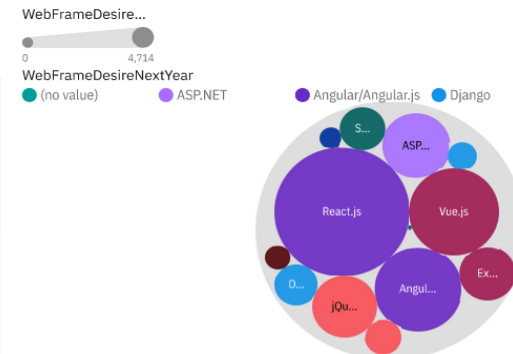


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Platforms Desired Next Year



Top 10 WebFrames Desired Next Year

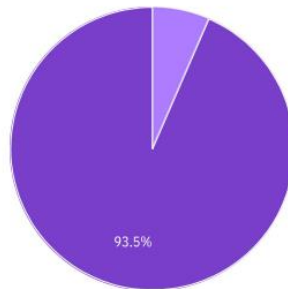


DASHBOARD TAB 3

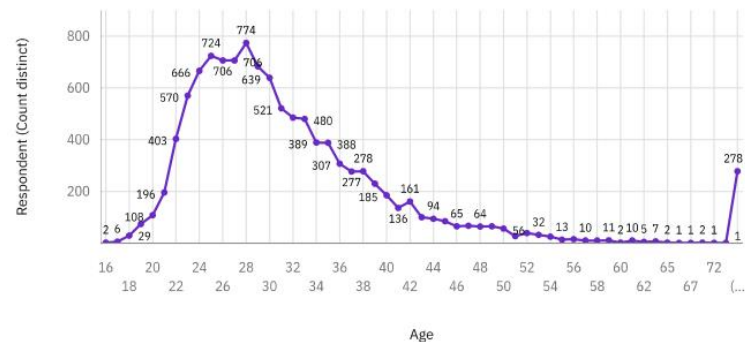
Demographics

Respondent Classified by Gender

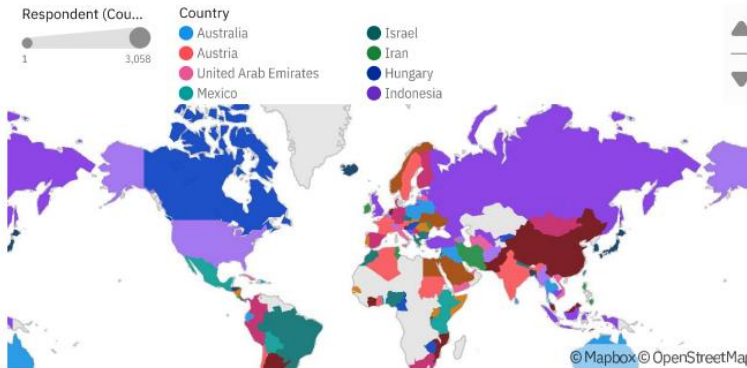
Gender
Woman Man



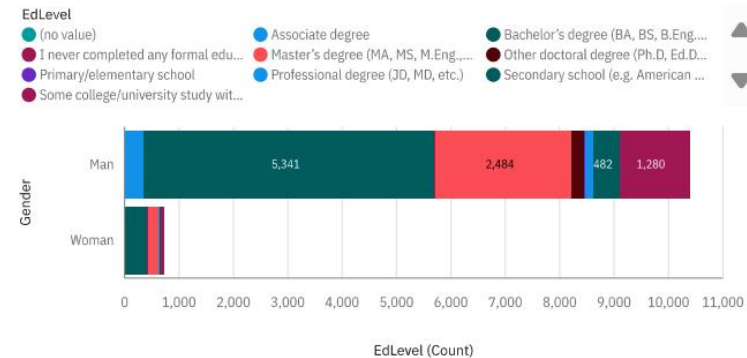
Respondent Count by Age



Respondent Count for Countries



Respondent by Gender and Education Level



DISCUSSION



OVERALL FINDINGS & IMPLICATIONS

Findings

- Windows is the most popular platform used. Followed by Linux, Docker and AWS.
- JQuery is most popular web framework used now. However React.js is the most desire for next year.
- More than 90% of developers are Males
- Vast majority of the developers have a either a Bachelors or a Masters Degree

Implications

- Linux, Docker and AWS are also the most desired platform for next year meaning they are high in demand.
- React.js is gaining popularity when it comes to web framework.
- Software developer profession is clearly not popular choice for women.
- Most of the software developer jobs require a formal degree.

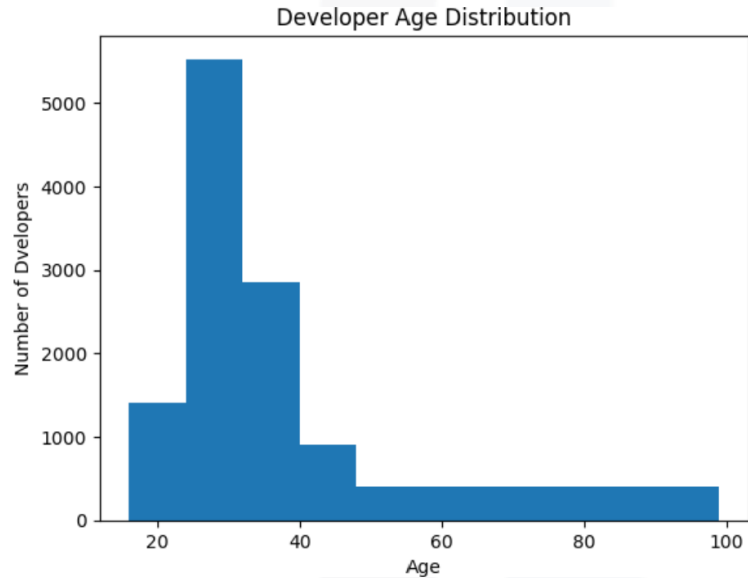
CONCLUSION



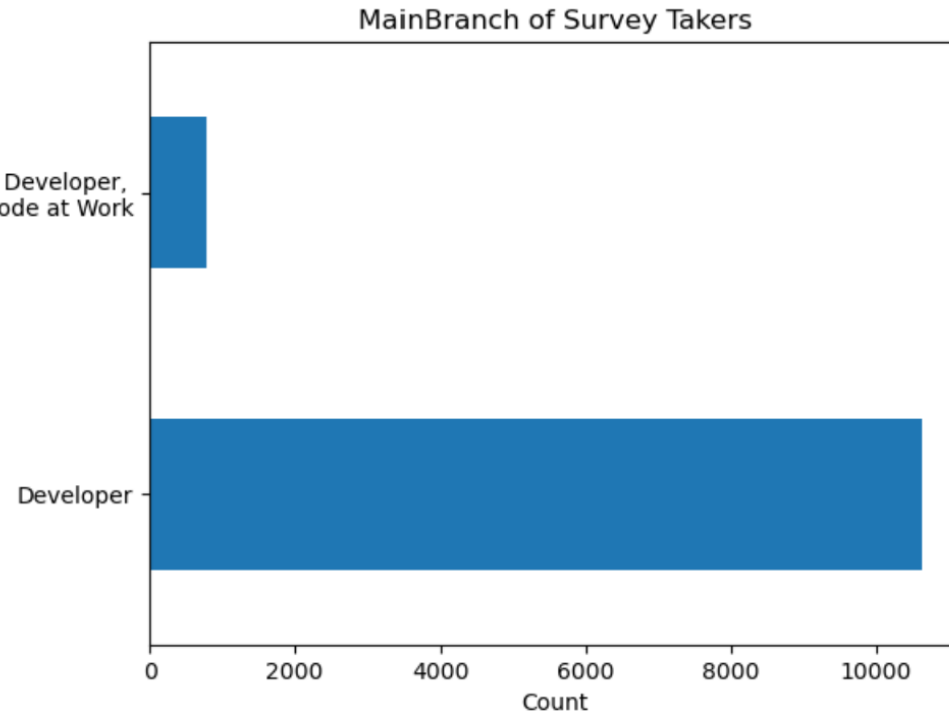
- Javascript and Html are the most popular languages used
- Databases such as PostgreSQL, MongoDB and Redis are being more popular
- Windows is the most widely used platform followed by Linux
- React.js web framework is gaining more popularity
- Far more male developers than female developers

APPENDIX

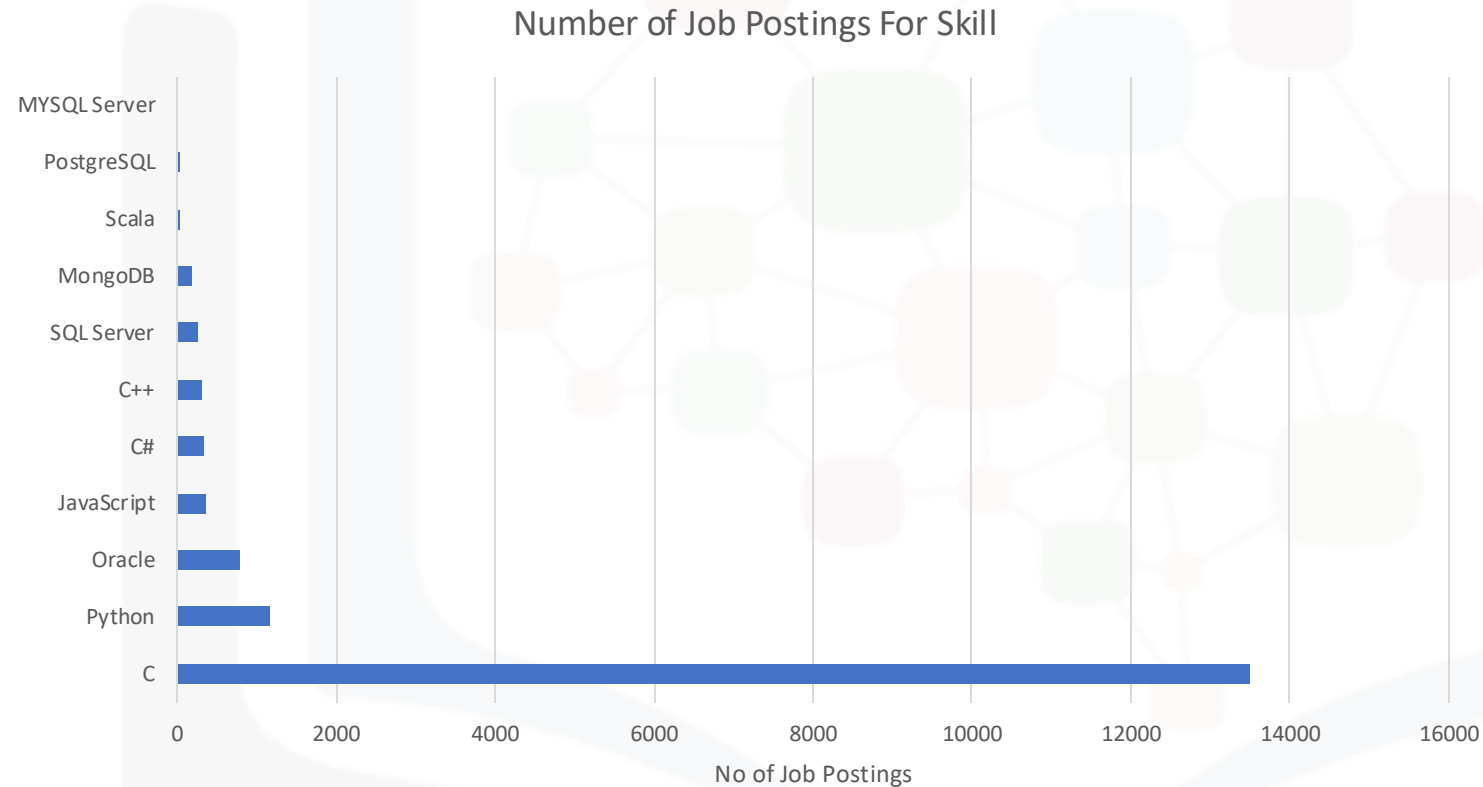
Age distribution of developers



Branch of Survey Takers



JOB POSTINGS



POPULAR LANGUAGES

